Hyperspectral Image Denoising via Robust Subspace Estimation and Group Sparsity Constraint

Xiyou Fu, *Member, IEEE*, Yujuan Guo, Meng Xu, *Member, IEEE*, and Sen Jia, *Senior Member, IEEE*

***Abstract*— Hyperspectral cameras capture electromagnetic information within hundreds of narrow spectral bands, producing hyperspectral images (HSIs) with the capability to accurately characterize the attribute information of objects. However, mixed noise induced by instrument and atmospheric effects hinders the interpretations and applications of the HSIs. In this article, we propose a novel subspace representation-based mixed noise removal method for HSIs via robust subspace estimation and weighted group sparsity constraint (RoSEGS). An outlier detection method is proposed to effectively detect sparse noise and replace the sparse noise with new estimates. A subspace estimation strategy, which is robust to mixed noise, is proposed. The subspace is first estimated after sparse noise detection and then optimized iteratively. In addition to the introduction of a state-of-the-art denoiser based on the plug-and-play technique to exploit self-similarity characteristics of the eigenimages, we impose a weighted group sparse regularization on the eigenimages to better promote the group sparsity of the spatial differences between the eigenimages, which further improves the denoising performance. We performed extensive experiments on two simulated and two real HSIs to fully demonstrate the effectiveness of the proposed method in comparison with seven state-of-the-art competitors.**

***Index Terms*— Group sparsity, hyperspectral image (HSI) denoising, noise removal, plug-and-play, subspace representation.**

Nomenclature

$\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$	3-dimensional tensor.
$\mathbf{X}$	Matrix.
$\mathbf{x}$	Vector.
x	Scalar.
order N	Number of dimensions in a tensor.
mode n	nth dimension of a tensor.
$\mathcal{X}(:,:,i)$	ith mode-3 slice of $\mathcal{X}$.
$\mathbf{X}_{(3)} \in \mathbb{R}^{I_3 \times (I_1 * I_2)}$	Mode-3 unfolding of $\mathcal{X}$.
$\|\mathcal{X}\|_F$	Frobenius norm of tensor $\mathcal{X}$: $\|\mathcal{X}\|_F = (\sum_{i_1,\ldots,i_N} \|x_{i_1,\ldots,i_N}\|^2)^{1/2}$.
$\|\mathcal{X}\|_{2,1}$	$\ell_{2,1}$ norm of tensor $\mathcal{X}$: $\|\mathcal{X}\|_{2,1} = \sum_{i_1,i_2} \|x_{i_1,i_2,:}\|_2$.
$\odot$	Componentwise multiplication.
$\times_k$	Mode-k tensor-matrix product.

Manuscript received 25 November 2022; revised 24 April 2023; accepted 8 May 2023. Date of publication 19 May 2023; date of current version 1 June 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62271327 and Grant 41971300; in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2022A1515011290 and Grant 2022A1515110076; and in part by the Shenzhen Science and Technology Program under Grant RCJC20221008092731042, Grant JCYJ20220818100206015, and Grant KQTD20200909113951005. *(Corresponding author: Sen Jia.)*

The authors are with the College of Computer Science and Software Engineering, the Guangdong-Hong Kong-Macau Joint Laboratory for Smart Cities, and the Key Laboratory for Geo-Environmental Monitoring of Coastal Zone, Ministry of Natural Resources, Shenzhen University, Shenzhen 518060, China (e-mail: fuxy0623@szu.edu.cn; guoyujuan@szu.edu.cn; m.xu@szu.edu.cn; senjia@szu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3277832

I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of continuous narrow spectral bands from visible wavelengths to near-infrared wavelengths, which greatly benefits the precise identification of the composed materials of the ground objects [1], [2]. In view of their capability to accurately characterize the attribute information of objects, HSIs play an important role in a wide range of tasks including dynamic monitoring of the environment [3], [4], land cover classification [5], [6], precision agriculture [7], and anomaly detection [8].

Different from multispectral imaging, hyperspectral cameras capture electromagnetic information within hundreds of narrow spectral bands. Unfortunately, due to the limitation of the hardware, the increase of spectral bands leads to fewer photons received by each band, leading to images with a lower signal-to-noise ratio (SNR). Apart from instrument noise, noise can also be introduced by atmospheric effects during the acquisition of the images [9]. The noise-induced can be a mixture, including Gaussian noise and sparse noise which is distributed sparsely in the spatial domain, such as "Salt and Pepper" noise and stripes [10]. The complex noise introduced not only severely affects the visualization of HSIs, but also limits the further application of HSIs in many areas [11], [12], such as classification accuracy [7] and anomaly detection accuracy [13], [14]. Therefore, restoring the latent clean images from the noisy observations is an important preprocessing task for HSIs before further applications [15], [16].

The denoising of HSIs is usually performed by imposing priors on the HSIs to exploit the different characteristics of HSIs and noise. Total variation (TV) can be used to promote piecewise smoothness in the spatial and spectral domains. For example, spectral–spatial adaptive hyperspectral TV (SSAHTV) [17] is proposed to promote localized step gradients within the image bands and align the "discontinuities" across the bands. Enhanced 3-DTV regularization method (E-3-DTV) [18] is proposed to calculate sparsity on the subspace obtained from gradient maps across the spectral bands of an HSI. Zhuang et al. [19] developed a unified TV-based mathematical model to restore the image corrupted by the cross-track illumination error and additive

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

Gaussian noise. Since HSI has a high spectral–spatial correlation, it is a feasible way to denoising an HSI based on the representation on a given basis. For example, Jiang and Hao [20] denoised the HSI on a Fourier basis. Rasti et al. [21] proposed a 3-D wavelet-based denoising method by using the ℓ_1 norm to promote sparsity on the wavelet coefficients. Different from Gaussian noise, HSI admits low rankness in the spectral domain [22], [23]. The significant difference allows us to minimize the rank of HSI to remove Gaussian noise in observations [24]. In practice, the nonconvex rank constraint is usually relaxed by minimizing the nuclear norm of HSIs, such as the low-rank matrix recovery (LRMR) method [25] and weighted sum of weighted tensor nuclear norm minimization (WSWTNNM) method [26]. Similar to natural images, HSI has the characteristic of self-similarity in the spatial domain. This feature can be utilized in the denoising problem by seeking similar patches and then estimating the patches by exploiting the low rankness or sparse representation via dictionary learning [27], achieving state-of-the-art denoising performance [28], [29]. Due to the intrinsic tensor form of HSI, many tensor-based methods were proposed for HSI denoising, including t-SVD decomposition [26], [30], Tucker decomposition [31], [32], CANDECOMP/PARAFAC (CP) decomposition [33], and tensor ring decomposition [34]. However, the performance of these methods generally heavily depends on the choice of rank, which is usually hard to determine in practice.

Recently, deep learning-based methods have been designed for HSI denoising. For example, Zhang et al. [35] proposed a spatial–spectral convolutional network, which simultaneously employs the spatial information and spectral information in fully cascaded multiscale convolutional neural network blocks. Chang et al. [36] proposed HSI-DeNet to jointly input the adjacent spectral bands and keep the spectral consistency of the HSIs. Zhang et al. [37] collaboratively modeled the nonindependent and identically distributed (i.i.d.) noise affecting the HSI and removed the noise within a deep spatio–spectral Bayesian posterior (DSSBP) framework. Although deep learning-based denoising methods could achieve state-of-the-art denoising results, the performance of deep learning-based methods highly lies in the availability of a large training dataset, which has unfortunately become a challenge for the application of deep learning-based denoising methods due to the lack of HSI pairs consisting of noisy-clean HSI pairs. One potential strategy is to incorporate deep learning-based denoisers well-trained with vast amounts of natural images, such as DnCNN [38] and FFDNet [39], into subspace representation-based denoising methods by taking advantage of the prevalent plug-and-play technique [40], [41].

Due to a large number of spectral bands, most HSI denoising methods suffer from high computational complexity. Fortunately, hyperspectral vectors can be approximated well with a subspace of a very low dimension [42], which have been proved and employed in many hyperspectral inverse problems, such as anomaly detection [13], image fusion [43], [44], spectral unmixing [45], and classification [10]. Following the idea of subspace representation, the denoising of an HSI can be formulated as the estimation of the subspace and the denoising of subspace representation coefficients, given the observed HSI. Subspace representation-based denoising methods, such as [28], [29], [46], and [47], can produce state-of-the-art results and are faster than many other methods. The key point of subspace representation-based denoising methods is the effective estimation of subspace and regularization of eigenimages, in particular in the case of mixed noise. In fact, there is still much space to improve the performance of the subspace representation-based mixed noise removal method by estimating the subspace robustly and taking the correlation among the eigenimages into consideration.

In this article, we propose a novel subspace representation-based mixed noise removal method for HSIs. The proposed method aims to reconstruct a clean HSI from degraded observations via robust subspace estimation and weighted group sparsity constraint (RoSEGS). Contributions of this article are given in detail in the following.

1) An outlier detection method is proposed to effectively detect sparse noise. By detecting the sparse noise samples and replacing them with new estimates, we can obtain a coarse image for a better estimation of the coarse subspace. Furthermore, the sparse noise mask is generated and embedded into the fidelity term of the objective function, ensuring that only reliable pixels are used to evaluate the similarity between the observed HSI and the denoised HSI.
2) A subspace estimation strategy that is robust to mixed noise is proposed. The coarse subspace is first obtained after the sparse noise detection using singular value decomposition (SVD), which can serve as a good initiative for the optimization of the objective function. The final accurate subspace is then obtained by optimizing the coarse subspace iteratively.
3) Except for the introduction of a state-of-the-art denoiser, BM3D, to exploit self-similarity characteristics of the eigenimages, we impose a weighted group sparse regularization on the eigenimages to better promote the group sparsity of the spatial differences between the eigenimages. The combination of the denoiser and the weighted group sparse regularization on the eigenimages further improves the denoising performance of the proposed method.

We organize the remainder of this article as follows. We introduce the related work of this article in Section II. In Section III, we introduce the proposed mixed noise removal method in detail. In Sections IV and V, we show and analyze the experimental results of the proposed method and the comparison methods using two simulated data and two real datasets. Last but not least, we make a conclusion of this article in Section VI.

II. Related Work

Subspace representation-based HSI denoising methods have been successfully exploited to remove Gaussian noise, such as FastHyDe [28] and NG-meet [29], and mixed noise, such as LRTF-DFR [48] and FastHyMix [46]. For Gaussian noise

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

removal, the key idea of the subspace representation-based methods lies in the effective regularizing of the subspace representation coefficients by taking advantage of the state-of-the-art denoisers, TV regularizers, and so on. For example, LRTF-DFR [48] and FGSLR [49] exploit TV regularization to take advantage of the piecewise smooth of the eigenimages. SNLRSF [50] and NG-meet [29] regularize the eigenimages by exploiting the low rankness of the nonlocal patches. The recently prevalent plug-and-play technique [40], [41] or named regularization by denoising (RED) framework [51], allows regularizing the by a state-of-the-art denoiser, achieving superior denoising performance, such as FastHyMix [46]. However, existing plug-and-play-based denoising methods only take advantage of the spatial information of each eigenimage without considering the correlation between eigenimages, which shows the potential to improve denoising performance.

When it comes to mixed noise, subspace representation-based HSI denoising methods come into another new challenge, i.e., the accurate and robust estimation of the subspace. When HSI is affected only by i.i.d. Gaussian noise, it is easy to estimate spectral subspace from a noise-corrupted HSI by simply performing SVD on the corrupted HSI matrix, which is the practice in FastHyDe [28] and NG-meet [29]. When HSI is affected by mixed noise including Gaussian noise, "Salt and Pepper" noise, and stripes, one way to obtain the spectral subspace is the joint estimation of the subspace and the corresponding subspace coefficients during the optimization of the objective function, as done in SNLRSF [50] and LRTF-DFR [48]. Estimation of the subspace and the corresponding coefficients of the HSI simultaneously may obtain an unsatisfactory subspace under the circumstance of serious mixed noise. To mitigate the negative effect of joint estimation, another strategy is to estimate the subspace before the optimization of the objective function, which is adopted in L1HyMixDe [52] and FastHyMix [46] method. However, when the HSI is affected by severe mixed noise, it is not easy to obtain an accurate estimation of subspace by only joint estimation of the subspace and the corresponding coefficients during the optimization of the objective function or only estimate the subspace before the optimization of the objective function.

Based on the related work introduced above, this article proposes an HSI mixed noise removal method based on the improvement of the two aspects: 1) the robust and accurate estimation of subspace and 2) effectively regularizing the representation coefficients. For the estimation of the subspace, we propose a subspace estimation method robust to severe mixed noise cases. For the regularization of subspace representation coefficients, we exploit the spatial self-similarity characteristic of the eigenimages by using a nonlocal image prior. In addition, we imposed a weighted group sparse regularizer on eigenimages to promote the group sparsity of the spatial difference images between each eigenimage, thus further improving the regularization of the representation coefficients.

III. Proposed Method

In this section, we will introduce the proposed denoising method in detail. Following the nomenclatures of [53], we first give the notations and their interpretations used in this article in Nomenclature.

A. Observation Model

Let $\mathcal{X} \in \mathbb{R}^{r\times c\times n_b}$ denote an underlying clean HSI, where r, c, and n_b are the number of rows, columns, and bands, respectively. Under the assumption that noise is additive, we can derive the observation model as

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

where $\mathcal{Y}, \mathcal{N} \in \mathbb{R}^{r\times c\times n_b}$ are the observed HSI and noise, respectively. In general, elements in $\mathcal{N}$ are composed of Gaussian noise, stripes, and "Salt and Pepper" noise.

The spectral bands of HSIs have a high correlation, which means that spectral vectors (i.e., mode-3 vectors) of an HSI $\mathcal{X}$ can be well approximated by vectors from a low-dimensional subspace [13], [28], [42]. We used two representative materials (i.e., grass and building) from the cropped Washington DC Mall image as an example. From Fig. 1(a) and (c), we can clearly observe that the two extracted spectral vectors can be well approximated by vectors living in a subspace with a dimension of 10. From Fig. 1(d), we can see that the first 100 bands are highly correlated. Due to the high spectral correlation, the autocorrelation matrix can be approximated by a low-rank matrix.

Based on the subspace representation of the HSI, we can write

$$\mathcal{X} = \mathcal{Z} \times_3 \mathbf{E} + \Xi \tag{2}$$

with $\mathbf{E} \in \mathbb{R}^{n_b\times p}$ ($p \ll n_b$), $\mathcal{Z} \in \mathbb{R}^{r\times c\times p}$, and $\Xi \in \mathbb{R}^{r\times c\times n_b}$. Matrix $\mathbf{E}$ holds an orthogonal subspace basis for an HSI. Hereafter, the mode-3 slices of $\mathcal{Z}$ are also called eigenimages. $\mathcal{Z} \in \mathbb{R}^{r\times c\times p}$ is a representation tensor whose entries are coefficients correspond to $\mathbf{E}$. Tensor Ξ represents residuals of low dimensional representation of $\mathcal{X}$. When SVD is adopted, most of the image variation can be compactly represented in a low dimensional subspace and the residual tensor contains a very small amount of image energy. Thus, we omit Ξ when reconstructing the image, that is, $\mathcal{X} = \mathcal{Z} \times_3 \mathbf{E}$. The observation model (1) can be written as

$$\mathcal{Y} = \mathcal{Z}\times_3\mathbf{E} + \mathcal{N}. \tag{3}$$

Subspace representation of HSIs has been applied to solve many hyperspectral inverse problems and has shown satisfactory performance, such as HSI anomaly detection [13], HSI super-resolution [54], and HSI unmixing [55]. However, for mixed noise removal of HSIs, the difficulties of utilizing subspace representation lie in the accurate estimation of the subspace from observations contaminated by not only Gaussian noise but also sparse noise, and the effective regularization of representation coefficient $\mathcal{Z}$ by exploiting the information at different the modes. These challenges motivate us to propose a subspace representation mixed noise removal method by estimating subspace robustly and regularizing the eigenimages effectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

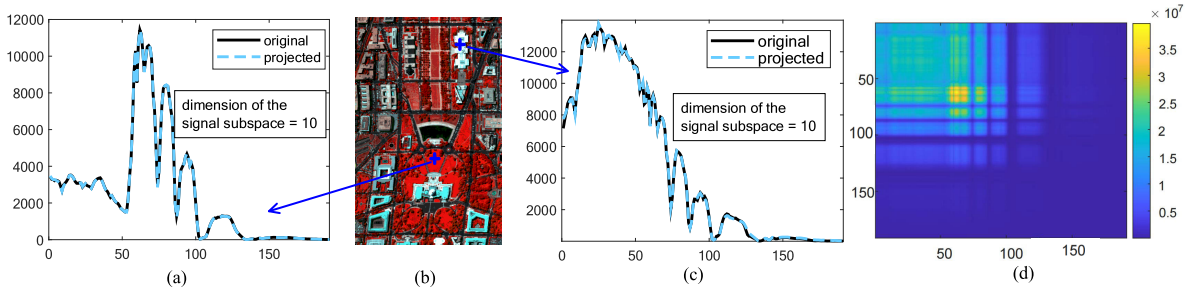


Fig. 1. Subspace representation of two representative spectral vectors and spectral correlation in Washington DC Mall image. (a) Spectral of grass. (b) Washington DC Mall image. (c) Spectral of building. (d) Autocorrelation matrix of spectral vectors.

B. Sparse Noise Detection and Coarse Subspace Estimation

1) Sparse Noise Detection: Since the noise in HSIs is normally composed of Gaussian noise and sparse noise such as stripes, and "Salt and Pepper" noise, we first proposed a sparse noise detection method to detect sparse noise. A spectral vector can be viewed as a 1-D signal. Thus, the detection of sparse noise is similar to outlier detection from a 1-D signal. The sparse noise can be viewed as outliers and the remaining pixels are termed as reliable samples. After the outlier detection, we can generate a sparse noise mask tensor, whose entries corresponding to the detected sparse noise are zero and the remaining entries are one. The advantage of sparse noise detection is two folds. First, by detecting the sparse noise samples and replacing them with new estimates, we can generate a coarse image for better estimation of the subspace. Second, by embedding the sparse noise mask into the fidelity term in the proposed model, only the most reliable pixels are used to guarantee the similarity between the observed HSI and the denoised HSI. A simple yet effective outlier removal step is introduced to detect sparse noise in the spectral vectors below.

Let $\mathbf{y} = [y_1, y_2, \ldots, y_{n_b}]^T \in \mathbb{R}^{n_b}$ denote a spectral vector of the observed HSI, and q denotes the length of the sliding window, which is fixed to 3 in this article. The median for a sample y_i within a sliding window of q can be written as

$$m_i = \text{median}\big(y_{i-q}, y_{i-q+1}, y_{i-q+2}, \ldots, y_i, \ldots \\ y_{i+q-2}, y_{i+q-1}, y_{i+q}\big) \quad (4)$$

where median$(\cdot)$ is the function that returns the median values. The median absolute deviation (MAD) estimates for a sample y_i is defined as

$$\text{MAD} = \text{median}\big(|y_{i-q} - m_i|, \ldots, |y_{i+q} - m_i|\big). \quad (5)$$

The standard deviation, $\sigma_{i,\text{MAD}}$, is estimated by multiplying the local MAD by a scale factor as

$$\sigma_{i,\text{MAD}} = \kappa \text{MAD} \quad (6)$$

where $\kappa = 1/((2)^{1/2}\text{erf}^{-1}(1/2)) \approx 1.4826$. We then follow the three-sigma rule of statistics which is robust against outliers to detect the sparse noise. Specifically, if a sample y_i satisfies

$$|y_i - m_i| > 3\sigma_{i,\text{MAD}} \quad (7)$$

then y_i is declared as an outlier. The sparse noise mask $\mathcal{M} \in \mathbb{R}^{r\times c\times n_b}$ can be obtained by setting all samples detected as outliers to 0 and the remaining samples to 1.

2) Coarse Subspace Estimation: For Gaussian distributed noise that existed in an observed HSI, we can estimate the subspace basis directly from the noisy observation by performing SVD on the noisy image matrix, because the addition of Gaussian noise does not change the order of singular values and consequently does not affect the estimation of subspace. Mathematically, subspace $\mathbf{E} = \mathbf{U}(:, 1:p)$, where $\{\mathbf{U}, \Sigma, \mathbf{V}\} = \text{SVD}(\mathbf{Y}_{(3)})$, $\mathbf{U} \in \mathbb{R}^{n_b\times n_b}$, and $\mathbf{V} \in \mathbb{R}^{p\times p}$ are orthogonal matrices, Σ is a diagonal matrix whose diagonal elements are singular values ranked by nonincreasing magnitude. However, when noise is composed of Gaussian noise and sparse noise, such as stripe and "Salt and Pepper" noise, directly performing SVD on the noisy image matrix is not an optimal way to learn spectral subspace, because the existence of sparse noise increases the variance of each dimension nonuniformly, which will severely impact the accuracy of the estimated subspace, in particular in the terribly noisy case.

To generate a coarse image without severe sparse noise for subspace estimation, the samples detected as outliers are replaced with m_i as

$$\tilde{y}_i = \begin{cases} m_i, & \text{if } |y_i - m_i| > n_\sigma \sigma_{i,\text{MAD}} \\ y_i, & \text{otherwise.} \end{cases} \quad (8)$$

The new estimates may not be the exact spectral values of the clean HSIs. However, by replacing the detected samples in the observed image with new estimates closer to the truth, we can obtain a coarse image without severe sparse noise, which will greatly benefit the estimation of subspace basis.

After replacing all the detected samples, we can obtain the coarse image $\widetilde{\mathcal{Y}}$. Then, we can easily estimate the subspace from $\widetilde{\mathbf{Y}}$, which contains approximately Gaussian noise. Specifically, the subspace can be obtained by performing SVD on $\widetilde{\mathbf{Y}}$, and extracting the first p left-singular vectors

$$\widehat{\mathbf{E}} = \widetilde{\mathbf{U}}(:, 1:p) \quad (9)$$

where $\{\widetilde{\mathbf{U}}, \widetilde{\Sigma}, \widetilde{\mathbf{V}}\} = \text{SVD}(\widetilde{\mathbf{Y}}_{(3)})$, $\widetilde{\mathbf{U}} \in \mathbb{R}^{n_b\times n_b}$ and $\widetilde{\mathbf{V}} \in \mathbb{R}^{p\times p}$ are orthogonal matrices, $\widetilde{\Sigma}$ is a diagonal matrix. The dimension of subspace, p, can be automatically estimated using HySime method [56]. We remark that the subspace basis estimated from the coarse image is still not an accurate subspace. However, the subspace basis estimated from the coarse image can serve as a good initiation for the following optimization of the objective function, which will be a great help for estimating the accurate subspace basis robustly even in the terribly noisy case.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

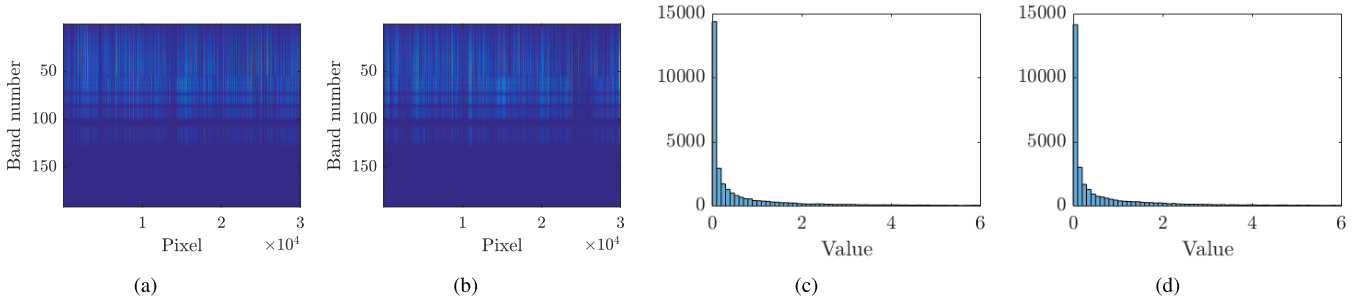


Fig. 2. Group sparse characteristics of the spatial differences images in a subimage of Washington DC Mall data. (a) Spatial horizontal difference image. (b) Spatial vertical difference image. (c) Histogram of the ℓ_2-norm values for all column vectors of (a). (d) Histogram of the ℓ_2-norm values for all column vectors of (b).

C. Self-Similarity and Group Sparsity in Eigenimages

Like other natural images, HSIs are images of the real world, thus they generally contain many similar patches at different locations and scales, which is called self-similarity. Since the bands of an HSI are self-similar and each band has the same pattern of self-similarity, then its corresponding eigenimages also have the property of self-similarity [28]. The exploitation of self-similarity in the eigenimages achieves state-of-the-art denoising performance, such as FastHyDe [28] and NG-meet [29].

In addition, since different bands present the same scene of the ground, different bands in HSIs should share the same patterns of piecewise smooth structure. In other words, the differencing images at two spatial directions across the entire bands should have group sparse characteristics along the spectral dimension. We use a subimage of Washington DC Mall data as an example to illustrate the group sparse characteristics of the spatial differences images. The spatial horizontal and vertical differences images unfolding along mode-3 are shown in Fig. 2(a) and (b), respectively. In Fig. 2(a) and (b), group sparsity patterns along the spectral mode can be clearly observed. We present the histogram of the ℓ_2-norm values for all column vectors of Fig. 2(a) and (b) in (c) and (d), respectively. It can be clearly seen from Fig. 2(c) and (d) that most of the ℓ_2-norm values tend to be zero, indicating the group sparsity of the spatial difference images in the HSI. Since eigenimages of the HSI share the same spatial patterns as the original HSI, it can be easily known that the spatial difference images of the eigenimages also have group sparsity along the third mode.

Thus, in this article, except for the nonlocal prior to exploiting self-similarity of the eigenimages, we introduce a weighted group sparse regularization [57] to promote the group sparsity of the spatial difference images along the third mode in the eigenimages as

$$\mathcal{J}(\mathcal{Z}) = ||W \odot D\mathcal{Z}||_{2,1} \tag{10}$$

where D is the two differential operators in two spatial dimensions D_x and D_y, and W is the weight operator in two spatial dimensions W_x and W_y. The two differential operators are calculated as follows:

$$\begin{cases} D_x\mathcal{Z}(i,j,k) = \mathcal{Z}(i,j+1,k) - \mathcal{Z}(i,j,k) \\ D_y\mathcal{Z}(i,j,k) = \mathcal{Z}(i+1,j,k) - \mathcal{Z}(i,j,k). \end{cases} \tag{11}$$

The weighted group sparse regularization is formulated as

$$\begin{aligned} \|W \odot D\mathcal{Z}\|_{2,1} = &\sum_{i=1}^{m}\sum_{j=1}^{n} W_x(i,j,:)\|D_x\mathcal{Z}(i,j,:)\|_2 \\ &+ \sum_{i=1}^{m}\sum_{j=1}^{n} W_y(i,j,:)\left\|D_y\mathcal{Z}(i,j,:)\right\|_2. \end{aligned} \tag{12}$$

D. Objective Function and Optimization

In this article, to mitigate the effectiveness of sparse noise on noise removal, First, we embed the estimated sparse noise mask into the fidelity term of the proposed model. Then, we optimize the initial subspace basis in an iterative way to obtain the accurate subspace basis. Finally, to effectively learn the eigenimages, we jointly use the nonlocal prior to exploit the spatial self-similarity of the eigenimages and a weighted group sparsity regularizer to promote the group sparsity along the mode-3 of the eigenimages. The objective function of the proposed method is written as

$$\min_{\widehat{\mathbf{E}},\mathcal{Z}} \left\|\mathcal{M} \odot (\mathcal{Y} - \mathcal{Z}\times_3\widehat{\mathbf{E}})\right\|_F^2 + \lambda\phi(\mathcal{Z}) + \beta\|W \odot D\mathcal{Z}\|_{2,1} \tag{13}$$

where λ and β are regularization parameters. $\phi(\cdot)$ is the regularization of $\mathcal{Z}$ to take advantage of the nonlocal self-similarity of the eigenimages.

By introducing four variables, $\mathcal{V}_1$, $\mathcal{V}_2$, $\mathcal{R}$, and $\mathcal{Q}$, we can transform the objective function into a constrained optimization as

$$\begin{aligned} \min_{\widehat{\mathbf{E}},\mathcal{Z},\mathcal{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q}} \quad & \|\mathcal{M} \odot \mathcal{V}_1\|_F^2 + \lambda\phi(\mathcal{V}_2) + \beta\|W \odot \mathcal{R}\|_{2,1} \\ \text{s.t. } & \mathcal{V}_1 = \mathcal{Y} - \mathcal{Z}\times_3\widehat{\mathbf{E}}, \quad \mathcal{V}_2 = \mathcal{Z}, \ D\mathcal{Q} = \mathcal{R}, \ \mathcal{Q} = \mathcal{Z}. \end{aligned} \tag{14}$$

Next, we transformed the constrained problem into the following augmented Lagrangian function:

$$\begin{aligned} &L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i) \\ &= \|\mathcal{M} \odot \mathcal{V}_1\|_F^2 + \lambda\phi(\mathcal{V}_2) \\ &\quad + \beta\|W \odot \mathcal{R}\|_{2,1} + \mu_1\left\|\mathcal{V}_1 - \mathcal{Y} + \mathcal{Z}\times_3\widehat{\mathbf{E}} + \frac{\mathcal{G}_1}{2\mu_1}\right\|_F^2 \\ &\quad + \mu_2\left\|\mathcal{V}_2 - \mathcal{Z} + \frac{\mathcal{G}_2}{2\mu_2}\right\|_F^2 + \mu_3\left\|D\mathcal{Q} - \mathcal{R} + \frac{\mathcal{G}_3}{2\mu_3}\right\|_F^2 \end{aligned}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

$$+\mu_4\left\|\mathcal{Q}-\mathcal{Z}+\frac{\mathcal{G}_4}{2\mu_4}\right\|_F^2 \tag{15}$$

where $\mu_i (i=1,2,3,4)$ are the penalty parameters and $\mathcal{G}_i (i=1,2,3,4)$ are the Lagrange multipliers. By referring to the alternative method of multipliers (ADMM), we can split (15) into several small subproblems. Then, we can easily solve each subproblem alternatively.

1) Subproblem of $\mathcal{V}_1$: Subproblem of $\mathcal{V}_1$ can be solved by minimizing $L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)$ with respect to $\mathcal{V}_1$, such that

$$\begin{aligned}\mathcal{V}_1 &\in \arg\min_{\mathcal{V}_1} L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)\\ &= \arg\min_{\mathcal{V}_1} \|\mathcal{M}\odot\mathcal{V}_1\|_F^2+\mu_1\left\|\mathcal{V}_1-\mathcal{Y}+\mathcal{Z}\times_3\widehat{\mathbf{E}}\right\|_F^2.\end{aligned} \tag{16}$$

Since (16) is strongly convex, by setting (16) with respect to $\mathcal{V}_1$ to zero, we can obtain

$$\mathcal{M}\odot\mathcal{M}\odot\mathcal{V}_1+\mu_1\mathcal{V}_1=\mu_1\left(\mathcal{Y}-\mathcal{Z}\times_3\widehat{\mathbf{E}}\right) \tag{17}$$

whose vectorizing form is

$$\left(\frac{1}{\mu_1}\text{diag}(\mathcal{M}\odot\mathcal{M})+\mathcal{I}\right)\text{vec}(\mathcal{V}_1)=\text{vec}\left(\mathcal{Y}-\mathcal{Z}\times_3\widehat{\mathbf{E}}\right) \tag{18}$$

where $\text{diag}(\cdot)$ is a function to reshape a tensor into a diagonal matrix whose diagonal elements are the elements of the tensor, $\text{vec}(\cdot)$ is a function to vectorize a tensor into a vector, and $\mathcal{I}$ is the unit matrix. Equation (18) can be directly solved as

$$\mathcal{V}_1=\text{vec}^{-1}\left(\mathcal{T}^{-1}\text{vec}\left(\mathcal{Y}-\mathcal{Z}\times_3\widehat{\mathbf{E}}\right)\right) \tag{19}$$

where $\mathcal{T}=((1/\mu_1)\text{diag}(\mathcal{M}\odot\mathcal{M})+\mathcal{I})$, $\text{vec}^{-1}(\cdot)$ denotes the inverse operation of the vectorization, reshaping a vector to a tensor.

2) Subproblem of $\mathcal{V}_2$: Subproblem of $\mathcal{V}_2$ can be solved by minimizing $L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)$ with respect to $\mathcal{V}_2$ as

$$\begin{aligned}\mathcal{V}_2 &\in \arg\min_{\mathcal{V}_2} L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)\\ &= \arg\min_{\mathcal{V}_2} \lambda\phi(\mathcal{V}_2)+\mu_2\left\|\mathcal{V}_2-\mathcal{Z}+\frac{\mathcal{G}_2}{2\mu_2}\right\|_F^2.\end{aligned} \tag{20}$$

Here, we would like to make use of a recent plug-and-play prior to regularizing the eigenimages rather than conceiving a hand-crafted regularization $\phi(\mathcal{V}_2)$ [40], [41], [58]. According to the plug-and-play framework, we can solve problem (20) by plugging a denoising operator as follows:

$$\mathcal{V}_2\leftarrow\text{Denoiser}\left(\mathcal{Z}-\frac{\mathcal{G}_2}{2\mu_2},\frac{\lambda}{2\mu_2}\right) \tag{21}$$

where $\text{Denoiser}(\cdot)$ is a plugged denoiser to regularize the eigenimages. In this article, we plug the state-of-the-art denoiser, BM3D, to effectively take advantage of the spatial nonlocal self-similarity to denoise each eigenimage.

3) Subproblem of $\mathcal{Q}$: Subproblem of $\mathcal{Q}$ can be solved by minimizing $L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)$ with respect to $\mathcal{Q}$, such that

$$\begin{aligned}\mathcal{Q} &\in \arg\min_{\mathcal{Q}} L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)\\ &= \arg\min_{\mathcal{Q}} \mu_3\left\|D\mathcal{Q}-\mathcal{R}+\frac{\mathcal{G}_3}{2\mu_3}\right\|_F^2\\ &\quad+\mu_4\left\|\mathcal{Q}-\mathcal{Z}+\frac{\mathcal{G}_4}{2\mu_4}\right\|_F^2.\end{aligned} \tag{22}$$

Equation (22) is a least squared problem, which can be directly calculated by the linear equations system

$$(\mu_3 D^T D+\mu_4\mathcal{I})\mathcal{Q}=D^T(\mu_3\mathcal{R}-\mathcal{G}_3)+(\mu_4\mathcal{Z}-\mathcal{G}_4) \tag{23}$$

where $\mathcal{I}$ is the unit tensor. As we consider the periodic boundary condition for $\mathcal{Q}$, $D^T D$ possesses block circulant matrices with a circulant block structure. Thus, we can employ the fast Fourier transform (FFT) to efficiently solve (23) as follows [59]:

$$\mathcal{Q}=\text{ifftn}\left(\frac{\mathcal{B}}{\mu_3\mathbf{1}+\mu_4|\text{fftn(D)}|^2}\right) \tag{24}$$

where $\mathcal{B}=D^T(\mu_3\mathcal{R}-\mathcal{G}_3)+(\mu_4\mathcal{Z}-\mathcal{G}_4)$, $\text{fftn}(\cdot)$ and $\text{ifftn}(\cdot)$ are the fast 3-D Fourier transform and its inverse transform, respectively. The division operation is the elementwise division and $|\cdot|^2$ is the elementwise square.

4) Subproblem of $\mathcal{R}$: Subproblem of $\mathcal{R}$ can be solved by minimizing $L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)$ with respect to $\mathcal{R}$ as

$$\begin{aligned}\mathcal{R} &\in \arg\min_{\mathcal{R}} L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)\\ &= \arg\min_{\mathcal{R}} \mu_3\left\|D\mathcal{Q}-\mathcal{R}+\frac{\mathcal{G}_3}{2\mu_3}\right\|_F^2\\ &\quad+\beta\|W\odot\mathcal{R}\|_{2,1}.\end{aligned} \tag{25}$$

Let $\mathcal{C}=D\mathcal{Q}+(\mathcal{G}_3/2\mu_3)$, we can easily obtain the closed-form solution of the tube fibers of $\mathcal{R}$ by using the following calculations:

$$\mathcal{R}(i,j,:)=\text{shrink}_{2,1}\left(\mathcal{C}(i,j,:),|W(i,j,:)|\cdot\frac{\beta}{2\mu_3}\right) \tag{26}$$

where

$$\text{shrink}_{2,1}(\mathbf{x},\xi)=\begin{cases}\dfrac{\|\mathbf{x}\|_2-\xi}{\|\mathbf{x}\|_2}\mathbf{x}, & \text{if } \xi<\|\mathbf{x}\|_2\\ 0, & \text{otherwise}\end{cases} \tag{27}$$

$$W_x(i,j,:)=\frac{1}{\left\|\left(D_x\mathcal{Q}+\frac{\mathcal{G}_3}{2\mu_3}\right)(i,j,:)\right\|_2+\epsilon} \tag{28}$$

$$W_y(i,j,:)=\frac{1}{\left\|\left(D_y\mathcal{Q}+\frac{\mathcal{G}_3}{2\mu_3}\right)(i,j,:)\right\|_2+\epsilon} \tag{29}$$

and ϵ is a small constant to avoid singularities.

5) Subproblem of $\mathcal{Z}$: Subproblem of $\mathcal{Z}$ can be solved by minimizing $L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)$ with respect to $\mathcal{Z}$ as

$$\begin{aligned}\mathcal{Z} &\in \arg\min_{\mathcal{Z}} L(\widehat{\mathbf{E}}, \mathcal{Z}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)\\ &= \arg\min_{\mathcal{Z}} \mu_1\left\|\mathcal{V}_1-\mathcal{Y}+\mathcal{Z}\times_3\widehat{\mathbf{E}}+\frac{\mathcal{G}_1}{2\mu_1}\right\|_F^2\end{aligned}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

$$+\mu_2\left\|\mathcal{V}_2-\mathcal{Z}+\frac{\mathcal{G}_2}{2\mu_2}\right\|_F^2+\mu_4\left\|\mathcal{Q}-\mathcal{Z}+\frac{\mathcal{G}_4}{2\mu_4}\right\|_F^2. \tag{30}$$

Since (30) is strongly convex, by setting (30) with respect to $\mathcal{Z}$ to zero, we can obtain the closed-form solution of $\mathcal{Z}$ as

$$\mathcal{Z}=(\mu_1+\mu_2+\mu_4)^{-1}\left(\mu_1\left(-\mathcal{V}_1+\mathcal{Y}-\frac{\mathcal{G}_1}{2\mu_1}\right)\times_3\widehat{\mathbf{E}}^T+\mu_2\left(\mathcal{V}_2+\frac{\mathcal{G}_2}{2\mu_2}\right)+\mu_4\left(\mathcal{V}_4+\frac{\mathcal{G}_4}{2\mu_4}\right)\right). \tag{31}$$

6) Subproblem of $\widehat{\mathbf{E}}$: Subproblem of $\widehat{\mathbf{E}}$ can be solved by minimizing $L(\widehat{\mathbf{E}},\mathcal{Z},\mathcal{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i)$ with respect to $\widehat{\mathbf{E}}$ as

$$\begin{aligned}\widehat{\mathbf{E}}&\in\underset{\widehat{\mathbf{E}},\widehat{\mathbf{E}}^T\widehat{\mathbf{E}}=\mathbf{I}}{\arg\min}\,L(\widehat{\mathbf{E}},\mathcal{Z},\mathcal{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i)\\&=\underset{\widehat{\mathbf{E}},\widehat{\mathbf{E}}^T\widehat{\mathbf{E}}=\mathbf{I}}{\arg\min}\,\mu_1\left\|\mathcal{V}_1-\mathcal{Y}+\mathcal{Z}\times_3\widehat{\mathbf{E}}+\frac{\mathcal{G}_1}{2\mu_1}\right\|_F^2.\end{aligned} \tag{32}$$

The close-form solution of (32) is written as

$$\widehat{\mathbf{E}}=L(\xi)R(\xi)^T \tag{33}$$

where $L(\xi)$ and $R(\xi)$ are the left and right singular matrices of the matrix $\xi=-(\mathbf{V}_{1(3)}-\mathbf{Y}_{(3)}+(\mathbf{G}_{1(3)})/(2\mu_1))\mathbf{Z}_{(3)}^T$, respectively.

7) Updating Multipliers and the Penalty Parameter: The Lagrange multipliers $\mathcal{G}_i(i=1,2,3,4)$ and penalty parameter μ_2 are updated as follows:

$$\begin{cases}\mathcal{G}_1=\mathcal{G}_1+2\mu_1(\mathcal{V}_1-\mathcal{Y}+\mathcal{Z}\times_3\widehat{\mathbf{E}})\\\mathcal{G}_2=\mathcal{G}_2+2\mu_2(\mathcal{V}_2-\mathcal{Z})\\\mathcal{G}_3=\mathcal{G}_3+2\mu_3(D\mathcal{Q}-\mathcal{R})\\\mathcal{G}_4=\mathcal{G}_4+2\mu_4(\mathcal{Q}_4-\mathcal{Z})\\\mu_2=\min(\mu,\rho\mu_2)\end{cases} \tag{34}$$

where μ and $\rho>1$ denotes constant values.

Algorithm 1 Optimization of the Objective Function for the Proposed **RoSEGS** Denoising Method

1: **Input:** The noised HSI $\mathcal{Y}$, regularization parameters λ and β.
2: **Initialize:** μ_i, $\mathcal{V}_1$, $\mathcal{V}_2$, $\mathcal{Q}$, $\mathcal{R}$, $\mathcal{G}_i=0$.
3: Generate sparse noise mask $\mathcal{M}$ and estimate the coarse subspace $\widehat{\mathbf{E}}$.
4: **repeat**
5: Calculate $\mathcal{V}_1$ via (19)
6: Calculate $\mathcal{V}_2$ via (21)
7: Calculate $\mathcal{Q}$ via (24)
8: Calculate $\mathcal{R}$ via (26)
9: Calculate $\mathcal{Z}$ via (31)
10: Calculate $\widehat{\mathbf{E}}$ via (33)
11: Update $\mathcal{G}_i$ and μ_2 via (34)
12: **until** stopping criterion is satisfied.
13: **Output:** The restored HSI $\mathcal{X}=\mathcal{Z}\times_3\widehat{\mathbf{E}}$

We present the pseudocodes in Algorithm 1 to show how the proposed RoSEGS method is implemented to reduce mixed noise for an HSI. Given an HIS of size r (rows) $\times$ c (columns) $\times$ n_b (bands) with subspace dimension $p(p\ll n_b)$, the computational complexity of obtaining sparse noise mask $\mathcal{M}$ is $\mathcal{O}(rcn_bq\log q)$. The estimation of the coarse subspace $\widehat{\mathbf{E}}$ cost $\mathcal{O}(r^2c^2n_b)$. For each iteration, the updating of the $\mathcal{V}_1$ cost $\mathcal{O}(r^2c^2n_b^2)$. The updating of the $\mathcal{V}_2$ cost $\mathcal{O}(rcp+pd)$, where d represents the computational complexity of denoising an eigenimage using BM3D. FFT was employed to optimize the $\mathcal{Q}$ subproblem, which needs $\mathcal{O}(rcn_b\log(rcn_b))$. The updating of $\mathcal{R}$ subproblems using soft threshold shrinkage is $\mathcal{O}(rcn_b)$. Updating $\widehat{\mathbf{Z}}$ cost $\mathcal{O}(rcn_bp)$. Updating $\widehat{\mathbf{E}}$ cost $\mathcal{O}(rcn_bp+2n_bp^2)$. In summary, the total computation complexity of Algorithm 1 is $\mathcal{O}(r^2c^2n_b^2+rcp+pd+rcn_b\log(rcn_b)+rcn_bp+2n_bp^2)$ at every iteration.

IV. Experiments With Simulated Images

In this section, we conducted extensive experiments using two simulated datasets to fully evaluate the denoising performance of the proposed method in comparison with seven state-of-the-art competitors.

A. Simulation of Noisy Images

We used two commonly used HSI datasets, i.e., Pavia University data and Washington DC Mall data, for the simulation of the noisy images. For computation and memory convenience, we cropped subimages of the two datasets for experiments. The subimage of the Pavia University used has a size of 310 × 250 × 87. The subimage of the Washington DC Mall used has a size of 150 × 200 × 191. The pseudo images of the two HSIs are shown in Fig. 3(a) and (b), respectively.

We followed the same procedure in [46] to generate clean images and noisy images of the Pavia University data[1] and Washington DC Mall data.[2] Specifically, we first generated clean images by projecting spectral vectors of each image onto a subspace spanned by the eight most prominent eigenvectors. Then, the clean images were normalized into 0–1. We simulated noisy images by adding four different types of noises to the clean images, including Gaussian noise, "Salt and Pepper" noise, and stripe noise, as described in Cases 1–4.

Case 1 (Gaussian Non-i.i.d. Noise): The noise in ith pixel vector $\mathbf{n}_i\sim N(\mathbf{0},\mathbf{D}^2)$, where $\mathbf{D}$ is a diagonal matrix whose diagonal elements are sampled from a uniform distribution $U(0,0.01)$.

Case 2 (Gaussian Noise + Stripes): Gaussian noise (depicted in Case 1) and oblique stripe noise. The stripes randomly affect nearly 30% bands of the images and about 10% pixels of each band.

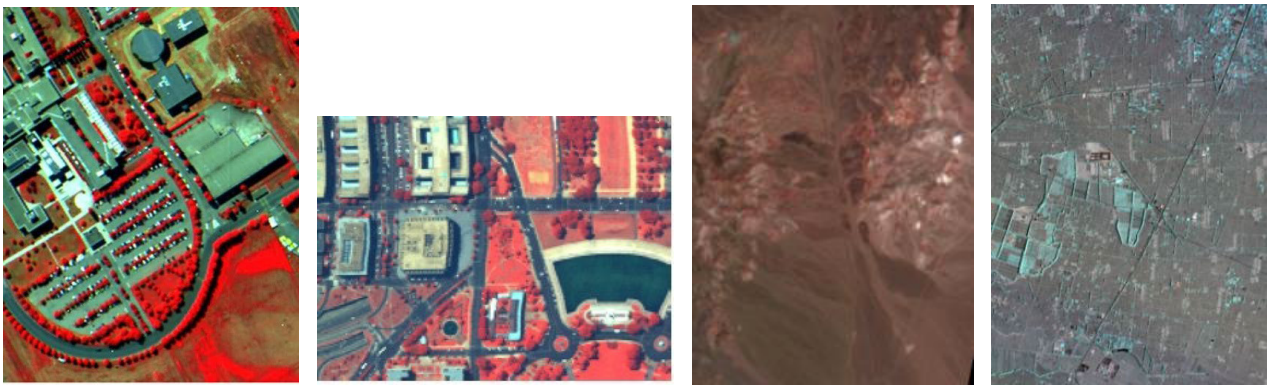
Case 3 (Gaussian Noise + "Salt and Pepper" Noise): Gaussian noise (depicted in Case 1) and "Salt and Pepper" noise. The noise density of the "Salt and Pepper" noise is 0.5%.

Case 4 (Gaussian Noise + Stripes + "Salt and Pepper" Noise): Gaussian noise (depicted in Case 1), random oblique stripes (depicted in Case 2), and "Salt and Pepper" noise (depicted in Case 3).

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[2] https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 3. Hyperspectral data used in the experiments. (a) Pavia University. (b) Washington DC Mall. (c) Hyperion Cuprite. (d) Tiangong-1 image.

TABLE I
DENOISING PERFORMANCE OF THE PROPOSED METHOD AND THE COMPETITORS ON TWO SIMULATED DATA

Cases	Indexes	Noisy	LRTM	NAILRMA	E-3DTV	FGSLR	LRTF-DFR	NMOG	FastHyMix	Proposed
Pavia University data										
Case 1	**MPSNR**	34.29	41.46	47.25	42.94	42.21	44.24	49.80	50.61	**52.14**
	MSSIM	0.8221	0.9783	0.9949	0.9914	0.9916	0.9914	0.9946	0.9980	**0.9982**
	MFSIM	0.9272	0.9908	0.9977	0.9955	0.9946	0.9956	0.9978	0.9991	**0.9992**
	Time (s)	-	58	142	45	167	127	303	**9**	242
Case 2	**MPSNR**	27.77	32.75	32.30	38.10	42.17	43.57	49.70	49.18	**52.05**
	MSSIM	0.7451	0.9182	0.9233	0.9739	0.9909	0.9906	0.9919	0.9976	**0.9981**
	MFSIM	0.8683	0.9471	0.9520	0.9824	0.9939	0.9954	0.9961	0.9989	**0.9992**
	Time (s)	-	57	138	46	841	125	291	**12**	236
Case 3	**MPSNR**	22.56	41.35	37.02	42.92	42.19	44.17	46.99	46.08	**51.95**
	MSSIM	0.6892	0.9778	0.9555	0.9913	0.9916	0.9911	0.9870	0.9948	**0.9981**
	MFSIM	0.8780	0.9906	0.9799	0.9954	0.9946	0.9955	0.9943	0.9977	**0.9991**
	Time (s)	-	58	85	45	188	125	290	**12**	234
Case 4	**MPSNR**	19.29	32.79	28.14	42.24	42.11	43.47	46.88	45.20	**51.43**
	MSSIM	0.6258	0.9167	0.8523	0.9902	0.9908	0.9904	0.9972	0.9951	**0.9980**
	MFSIM	0.8267	0.9467	0.9165	0.9948	0.9938	0.9953	0.9988	0.9976	**0.9991**
	Time (s)	-	58	134	45	856	127	291	**15**	233
Washington DC Mall data										
Case 1	**MPSNR**	19.13	30.35	37.22	36.03	34.26	37.14	40.46	42.02	**42.67**
	MSSIM	0.7781	0.9814	0.9962	0.9948	0.9930	0.9965	0.9980	0.9983	**0.9987**
	MFSIM	0.8634	0.9796	0.9931	0.9947	0.9935	0.9962	0.9978	0.9982	**0.9986**
	Time (s)	-	32	117	40	56	53	161	**6**	90
Case 2	**MPSNR**	12.52	21.48	20.60	30.98	34.90	34.31	40.48	38.60	**42.40**
	MSSIM	0.6851	0.9148	0.8439	0.9853	0.9937	0.9912	0.9979	0.9802	**0.9987**
	MFSIM	0.7973	0.9294	0.8936	0.9840	0.9951	0.9951	0.9978	0.9885	**0.9986**
	Time (s)	-	37	124	38	567	54	94	**10**	89
Case 3	**MPSNR**	7.29	30.50	25.08	36.54	34.26	36.98	40.47	38.38	**42.41**
	MSSIM	0.6421	0.9833	0.9643	0.9943	0.9930	0.9964	0.9980	0.9968	**0.9987**
	MFSIM	0.8052	0.9805	0.9713	0.9943	0.9935	0.9962	0.9978	0.9951	**0.9986**
	Time (s)	-	40	89	37	66	54	134	**11**	89
Case 4	**MPSNR**	3.84	21.33	15.79	24.99	34.90	34.21	40.33	33.47	**41.24**
	MSSIM	0.5676	0.9153	0.7946	0.9518	0.9937	0.9914	0.9979	0.9713	**0.9985**
	MFSIM	0.7508	0.9292	0.8651	0.9689	0.9951	0.9950	0.9979	0.9800	**0.9985**
	Time (s)	-	37	120	37	564	52	95	**13**	87

B. Competitors and Settings

We compare our proposed method with seven state-of-the-art HSI denoising methods to fully evaluate the performance. The seven state-of-the-art HSI denoising methods are LRMR [25], NAIRLMA [60], E-3DTV [18], FGSLR [49], NMOG [9], LRTF-DFR [48], and FastHyMix [46]. For all the competitors, the settings of parameters were obtained according to the corresponding literature or the original codes. The codes of this work will be available at https://github.com/FxyPd/RoSEGS.

To access the denoising results quantitatively, we employed mean peak SNR (MPSNR), mean structural similarity (MSSIM), and mean feature similarity (MFSIM) in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

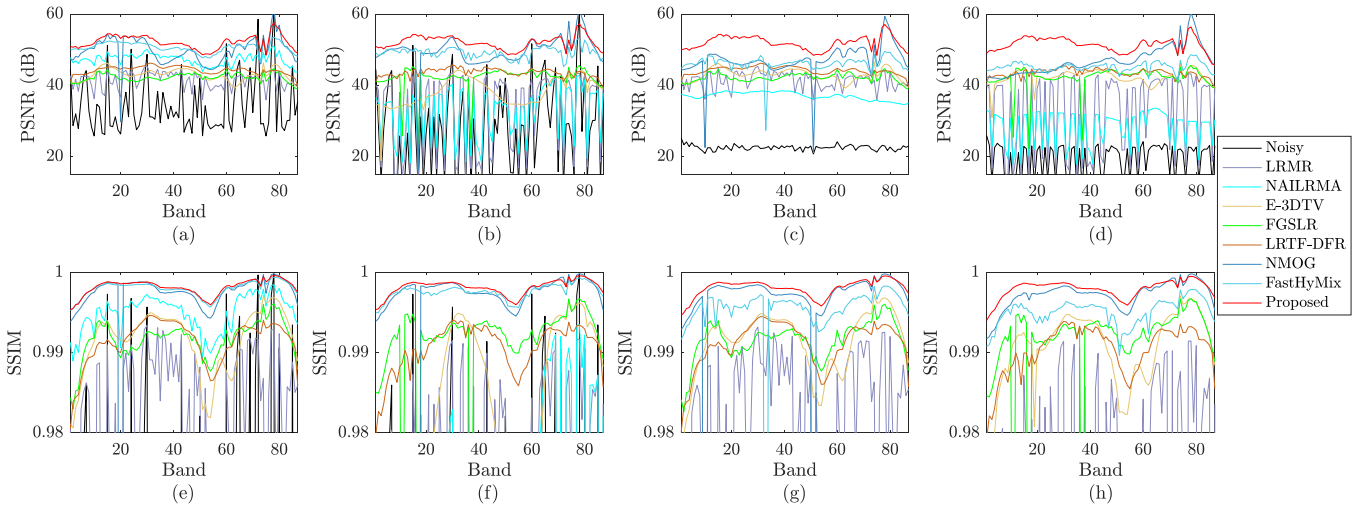


Fig. 4. Bandwise PSNR and SSIM values for the denoised Pavia University data. The first row and second row correspond to the PSNR and SSIM, respectively. The first column to the fourth row corresponds to Cases 1–4, respectively. (a) PSNR of Case 1. (b) PSNR of Case 2. (c) PSNR of Case 3. (d) PSNR of Case 4. (e) SSIM of Case 1. (f) SSIM of Case 2. (g) SSIM of Case 3. (h) SSIM of Case 4.

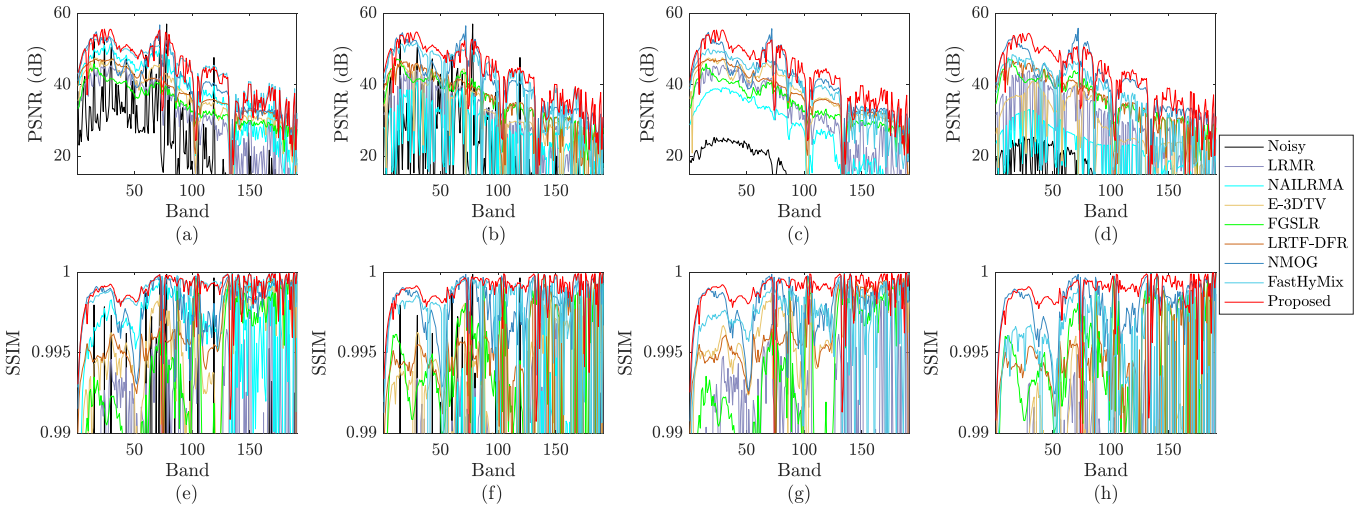


Fig. 5. Bandwise PSNR and SSIM values for the denoised Washington DC Mall data. The first row and second row correspond to the PSNR and SSIM, respectively. The first column to the fourth row corresponds to Cases 1–4, respectively. (a) PSNR of Case 1. (b) PSNR of Case 2. (c) PSNR of Case 3. (d) PSNR of Case 4. (e) SSIM of Case 1. (f) SSIM of Case 2. (g) SSIM of Case 3. (h) SSIM of Case 4.

assessment of the experimental results. We first calculated the PSNR, SSIM, and FSIM on each band of the two datasets, and then obtained the MPSNR, MSSIM, and MFSIM by averaging the PSNR, SSIM, and FSIM of all the bands on the two datasets.

C. Noise Removal Performance

The quantitative results of the proposed denoising method with LTMR [25], NAILRMA [60], E-3DTV [18], FGSLR [49], NMOG [9], LRTF-DFR [48], and FastHyMix [46], in terms of three matrices, i.e., MPSNR, MSSIM, MFSIM, and the running time, are represented in Table I. The best scores are marked in bold for clarity.

From Table I, we can see that when the HSIs are contaminated by only Gaussian noise as shown in Case 1, all the methods can remove noise to a certain degree. Among them, NAILRMA, LRTF-DFR, NMOG, FastHymix, and the proposed method produce the best results. However, when the HSIs are affected by mixed noise, LRMR, and NAILRMA fail to remove stripes effectively. Only NMOG and FastHymix can achieve comparable results with the proposed method. The proposed method yields uniformly the best results on both datasets in all cases, as shown in Table I. Among the comparisons, LRMR, NAILRMA, and FGSLR are low-rank and sparse representation-based methods. E-3DTV serves as a state-of-the-art TV-based method addressing mixed noise. Compared with these competitors, our method is a subspace representation-based method, which denoise the HSIs by effectively estimating the subspace and regularizing the subspace coefficients. From Table I, we can easily see the superiority of the proposed method over patch-based regularization, such as LRMR and NAILRMA, and TV regularization, such as E-3DTV.

The LRTF-DFR and FastHyMix are state-of-the-art subspace representation-based methods, which remove noise by regularizing the subspace coefficients of HSIs. The proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

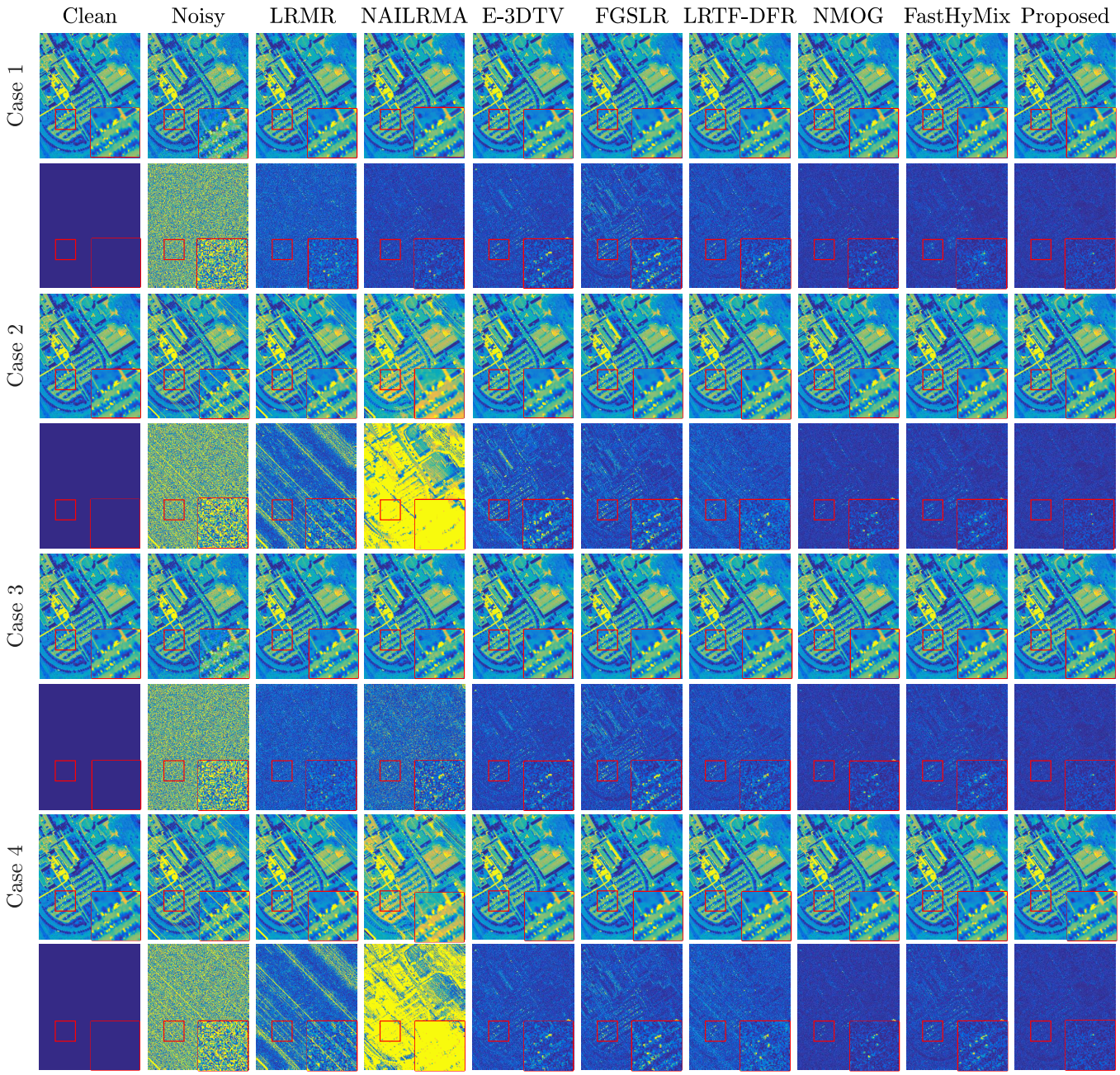


Fig. 6. Denoising results and the corresponding error images of the 40th band of the Pavia University data in four cases. The range of the error images is set to [0–0.03] for easy comparison of the difference.

method falls in line with LRTF-DFR and FastHyMix, which are also subspace representation-based methods by estimating the subspace vectors and imposing regularizations on representation coefficients. From Table I, we can clearly see that the proposed method achieves better denoising results than LRTF-DFR and FastHyMix. LRTF-DFR estimates the subspace and the eigenimages simultaneously, which may produce poor results when the image is contaminated by serous serious mixed noise. FastHyMix estimates a coarse subspace and uses the coarse subspace as the final subspace, which also limits the representation ability of the subspace. Compared with LRTF-DFR and FastHyMix, the proposed method elaborates an outlier removal operator to improve the estimation of the subspace. Furthermore, the coarse subspace is updated iteratively during the optimization to obtain a more accurate subspace. For the regularization of the eigenimages, the proposed method jointly takes advantage of the spatial information and spectral information of the eigenimages by incorporating a nonlocal prior and the weighted group sparsity regularizer. The two regularizations jointly contribute to the excellent performance of the proposed method in removing mixed noise from HSIs compared with the two subspace representation-based methods.

To quantitatively assess the denoising performance band by band, we present bandwise PSNR and SSIM values of the two datasets in Figs. 4 and 5, respectively. It can be easily seen

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

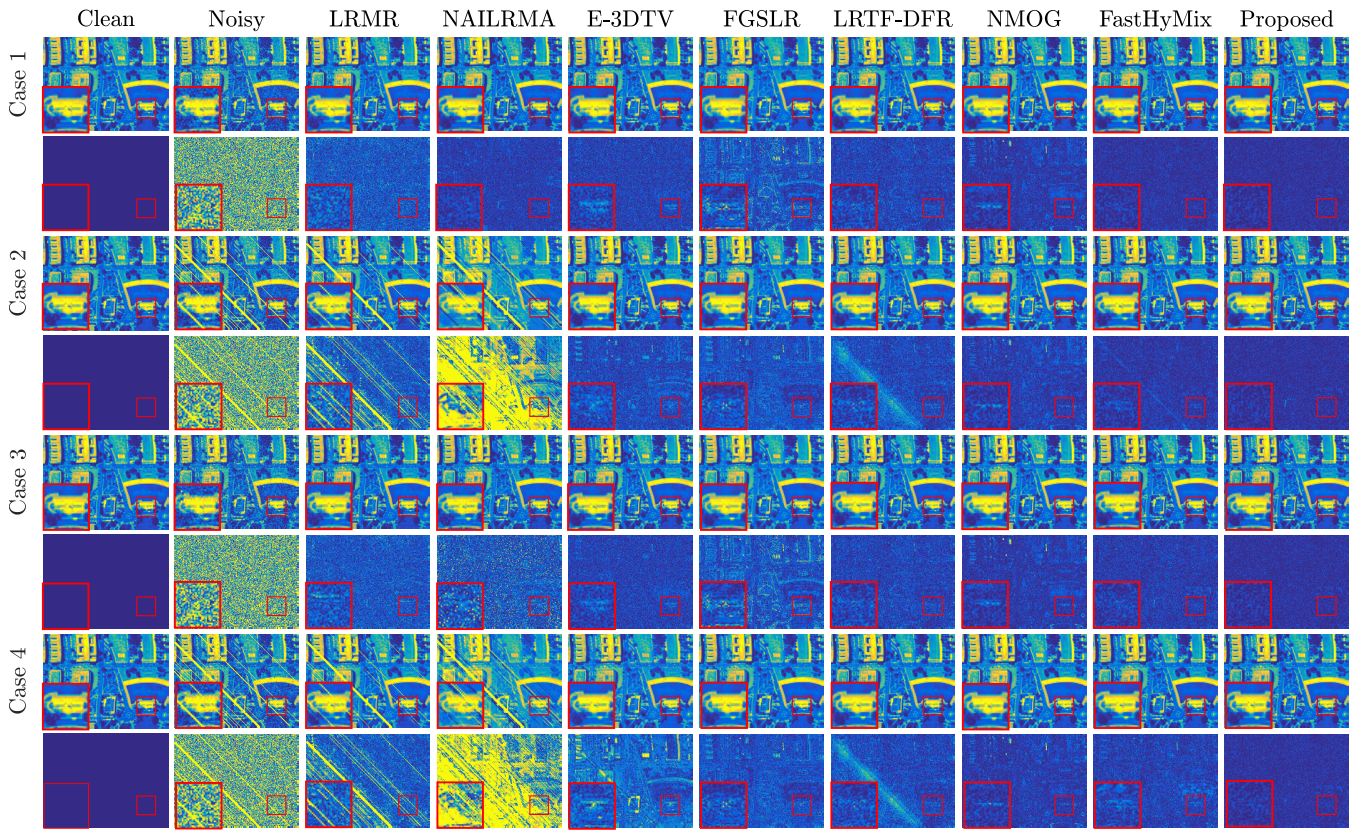


Fig. 7. Denoising results and the corresponding error images of the 44th band of the Washington DC Mall data in four cases. The range of the error images is set to [0–0.03] for easy comparison of the difference.

from Figs. 4 and 5 that when the noise is simple Gaussian noise as depicted in Case 1, all the methods are able to remove noise to some extent. However, the PSNR values of the proposed method are higher than other methods at most of the bands. In the cases of mixed noise, our proposed method can obtain more stable denoising results in comparison with other denoising methods. In particular, when the images are corrupted by strips or dead-lines, as shown in Cases 2 and 4, the proposed method outperforms other methods significantly. From the figures of SSIM, we can obviously observe that the proposed method can obtain higher PSNR values while preserving the spatial structure information of the images in comparison with the competitors.

For visual comparison, we present the 40th band of the Pavia University data and the corresponding error images in Fig. 6. Also, we present the 44th band of the Washington DC Mall data and the corresponding error images in 7, respectively. We use red rectangles to present and enlarge the chosen area for better visualization of the denoising results. The corresponding residual images were generated by calculating the absolute difference between the bands of the denoised images and the clean images. For Case 1 with only Gaussian noise and Case 3 with Gaussian noise and "Salt and Pepper" noise, all methods can reduce noise significantly on both datasets. As shown in Figs. 6 and 7, LRMR and NAILRMA fail to remove stripes effectively, leaving some stripes in the denoised images and some residuals in the error images. E-3DTV, FGSLR, and LRTF-DFR perform well on the Pavia University data, but obvious residuals can still be seen in the error images. NMOG, FastHyMix, and the proposed method yield visually comparable results as shown in the denoised bands of the images. However, from the corresponding error images, we can easily observe that the error images of the proposed method contain fewer residuals compared with NMOG and FastHyMix, as shown in Figs. 6 and 7. The reason why the proposed method achieves good denoising performance lies in the accurate subspace estimation under complex mixed noise conditions and the exploitation of the nonlocal self-similarity and group sparsity characteristics along the mode-3 of the eigenimages to effectively regularize eigenimages.

D. Analysis of Parameters

There are two regularization parameters, i.e., λ and β, to be tuned in the proposed method. The parameter λ controls the degree between the Gaussian noise removal and the information preservation. We present the MPSNR values of the proposed method with the varying regularization parameter λ on the two datasets in Fig. 8. It can be seen from Fig. 8 that the MPSNR values of the proposed method increase as λ increases from 0 to 0.6, and then drop slightly as λ continue to increase. Thus, we set λ to 0.6 in all the experiments.

The parameter β controls group sparse characteristics of the spatial differences of the eigenimages along the spectral dimension. To determine the value of β, we present the denoising performance of the proposed method on the Pavia

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

TABLE II
MPSNR VALUES OF THE ABLATION STUDY ON THE WASHINGTON DC MALL DATA

Cases	w/o coarse subspace estimation	w/o optimizing $\widehat{\mathbf{E}}$	$\lambda = 0$	$\beta = 0$	Proposed
Case 1	42.01	41.93	37.85	41.62	**42.67**
Case 2	35.49	41.49	37.41	41.95	**42.40**
Case 3	35.04	41.46	36.98	41.87	**42.41**
Case 4	34.24	40.04	36.61	40.95	**41.24**

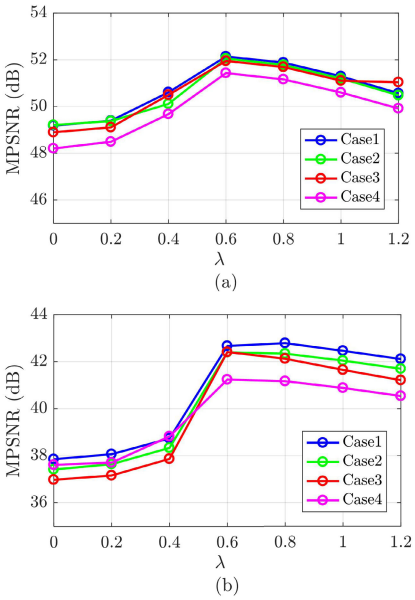


Fig. 8. PSNR values of the proposed method as a function of the parameter λ on the two datasets. (a) Pavia University. (b) Washington DC Mall.

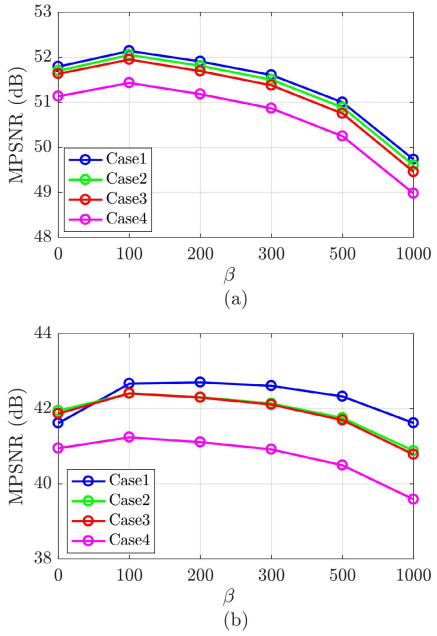


Fig. 9. PSNR values of the proposed method as a function of the parameter β on the two datasets. (a) Pavia University. (b) Washington DC Mall.

University dataset and Washington DC Mall dataset with the variation of β in Fig. 9. We can easily observe that when the value of β is set to 100, the proposed method achieves the highest MPSNR values on both datasets. The values of the MPSNR decrease as β continue to increase. Therefore, in this article, we set β to 100 in all experiments.

E. Ablation Study

To verify the contributions proposed in this article, we designed a series of ablation experiments on the Washington DC Mall data. The results of the ablation study in terms of MPSNR are presented in Table II. The MPSNR values of the proposed method are marked in bold for clarity. First, we performed an experiment without coarse subspace estimation to verify the influence of the coarse subspace estimation on the denoising results. The results without coarse subspace estimation are given in the first column of Table II. We can easily observe that in Case 1, i.e., the Gaussian noise, the MPSNR values without coarse subspace estimation are very close to the proposed method. However, when the HSI is affected by mixed noise, as described in Cases 1–4, the MPSNR drops significantly compared with the proposed method due to the poor estimation of the subspace under the mixed noise. Then, we present the experimental results without optimizing subspace $\widehat{\mathbf{E}}$ during the optimization of the objective function in the second column of Table II. It can be seen that the MPSNR values of the experiment without optimizing subspace $\widehat{\mathbf{E}}$ witness a decrease compared with the proposed method, which confirms the effectiveness of optimizing subspace $\widehat{\mathbf{E}}$ iteratively in improving denoising performance. The proposed method estimates a coarse subspace after the sparse noise detection and then optimizes the subspace $\widehat{\mathbf{E}}$ in an iterative way, which helps to obtain an accurate subspace and thus achieve better denoising results. Next, to evaluate the effectiveness of the plugged-in denoiser, i.e., BM3D, we present the experimental results of $\lambda = 0$ in the third column of Table II. Obviously, we can see that without the regularization of the plugged-in denoiser, the denoising results are significantly worse than the proposed method, suggesting the effectiveness of taking advantage of the spatial self-similarity in regularizing the eigenimages. Finally, we show the experimental results of $\beta = 0$ in the penultimate column of Table II to verify the effectiveness of the weighted group sparsity regularizer in regularizing the eigenimages. It can be easily observed from the penultimate column of Table II that all denoising results without the regularization of weighted group sparsity decrease compared with the proposed method. Except for the plugged-in denoiser, the introduction of weighted group sparsity to regularize the eigenimages further improves the denoising results of the proposed method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

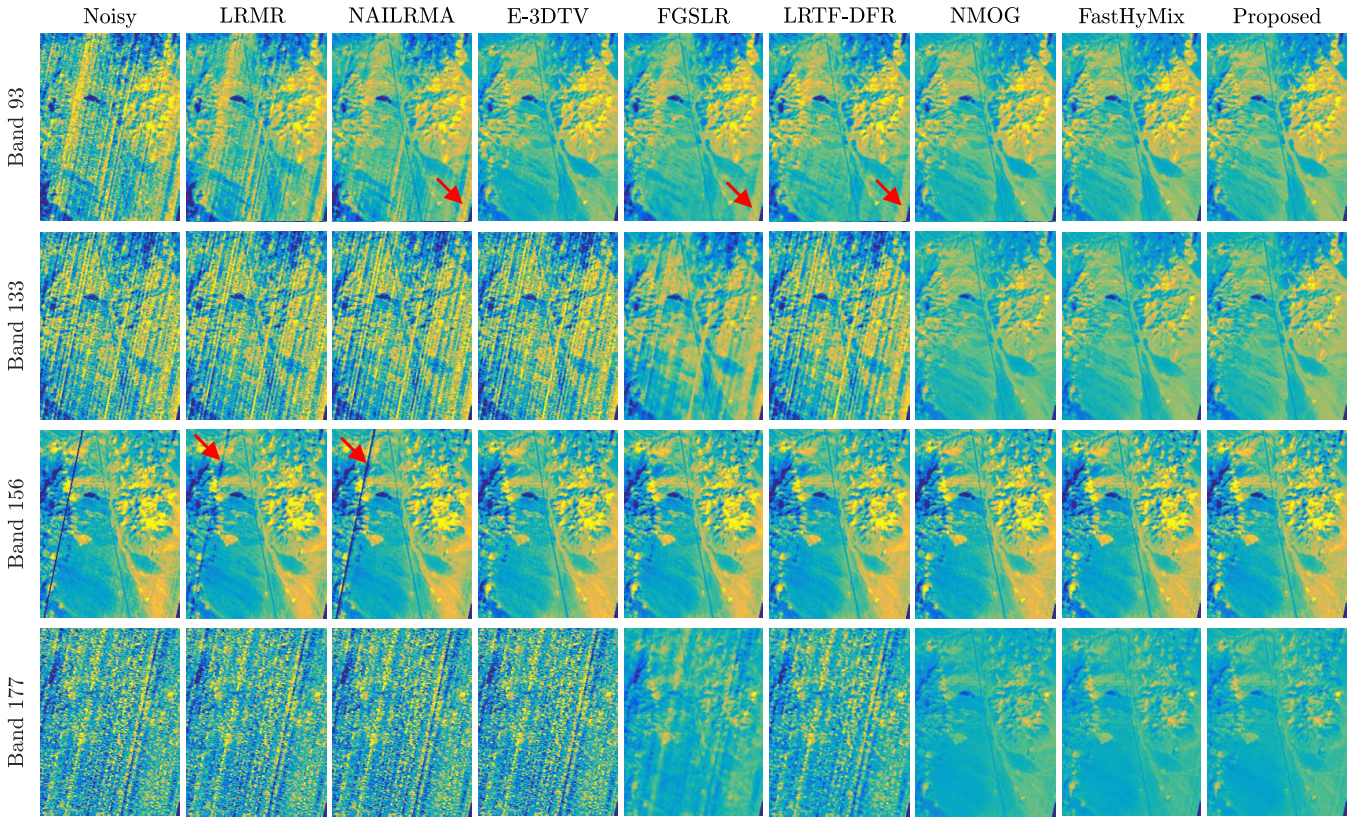


Fig. 10. Denoising results of the 93th, 133th 156th, and 177th bands of the Hyperion Cuprite data.

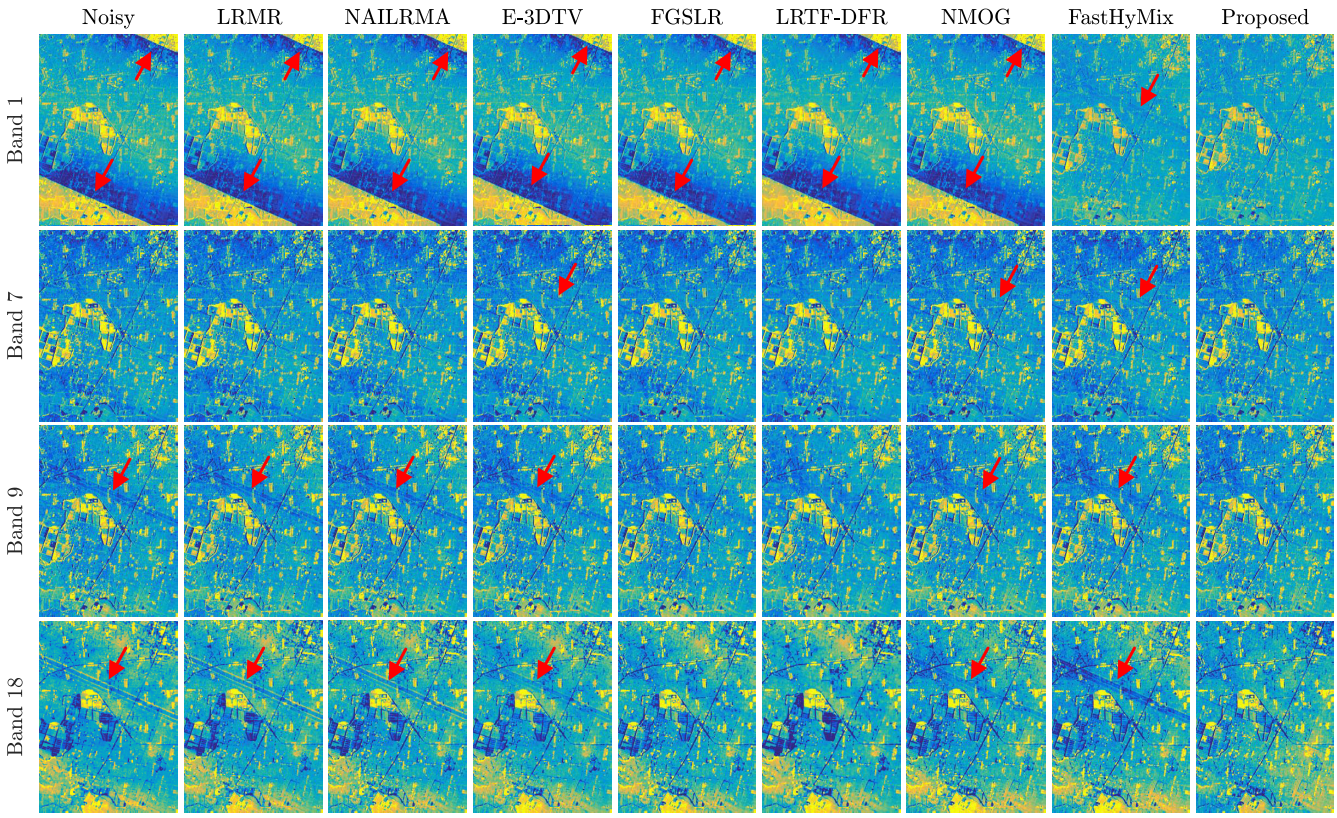


Fig. 11. Denoising results of the 1st, 7th, 9th, and 18th bands of the Tiangong-1 data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

V. Experimental Results for Real Images

In this section, we employ two real HSIs, i.e., the Hyperion Cuprite image[3] and the Tiangong-1 image [46], to evaluate the performance of the proposed method. The pseudoimages of the two HSIs are shown in Fig. 3(c) and (d), respectively.

A. Hyperion Cuprite Image

The Cuprite HSI was captured by the Hyperion sensor over Cuprite, NV, USA. It has 242 bands covering the wavelength range from 355 to 2577 nm with a spectral resolution of 10 nm. The spatial resolution of the image is 30 m. After removing bands with very low SNR, 177 spectral bands were left. A subimage of 240 × 178 pixels in size was cropped for the test. Four bands with different noises before and after the denoising are shown in Fig. 10. Among these bands, bands 93, 133, and 177 are affected by serious stripes, and band 156 was corrupted by a dead-line and slight stripes. From Fig. 10, we can see that LRMR and NAILRMA cannot remove the dead-line effectively in band 156, as marked by the red arrows. For bands 93, 133, and 177, which are contaminated by severe stripes, only NMOG, FastHymix, and the proposed method can effectively remove strips, achieving very similar good denoising performance.

B. Tiangong-1 Image

The Tiangong-1 data was obtained by a sensor placed in the Tiangong-1 satellite. The Tiangong-1 imager has a push broom scanner with 75 bands, whose nominal bandwidths are 23 nm, varying from 800 to 2500 nm. The scene used for testing was captured over an area of Qinghai, China, in May 2013. A subimage with 351 × 253 pixels in size was used in the experiment. We display four bands containing different noises before and after the denoising in Fig. 11. Among those bands, band 1 is contaminated by severe cross-track illumination errors. Bands 7, 9, and 18 are affected by obvious stripes. After denoising, we can see that only FastHymix and the proposed method can effectively remove the illumination errors in band 1, but FastHymix will induce some stripes, as indicated by the red arrow. For bands 7, 9, and 18, the competitors cannot remove the stripe completely, which has been indicated by the red arrows. On the contrary, the proposed method can effectively alleviate the stripe noise on all four bands. In all, our proposed method can obtain overall the best denoising result on the two real HSIs compared with the competitors.

VI. Conclusion

This article proposes a hyperspectral mixed noise removal method by improving the estimation of subspace and regularization of the representation coefficients. The proposed method introduces a subspace estimation method robust to severe mixed noise cases by first estimating a coarse subspace after sparse noise detection and then optimizing the subspace iteratively. For the regularization of subspace representation coefficients, the proposed method exploits the spatial self-similarity characteristic of the eigenimages by using a nonlocal image denoiser. Furthermore, the proposed method imposes a weighted group sparse regularizer on eigenimages to promote the group sparsity of the spatial difference images between each eigenimage. The combination of the denoiser and the weighted group sparse regularizer helps to improve the denoising performance of the proposed method. Two simulated and two real HSIs were used to evaluate the effectiveness of the proposed method. We compared the proposed denoising methods with seven state-of-the-art competitors. Extensive experimental results show the superiority of the proposed method for mixed noise removal compared with the competitors.

[3]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

References

[1] L. Zhuang, M. K. Ng, X. Fu, and J. M. Bioucas-Dias, "Hy-demosaicing: Hyperspectral blind reconstruction from spectral subsampling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515815.

[2] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.

[3] J. Bian et al., "Monitoring fractional green vegetation cover dynamics over a seasonally inundated alpine wetland using dense time series HJ-1A/B constellation images and an adaptive endmember selection LSMM model," *Remote Sens. Environ.*, vol. 197, pp. 98–114, Aug. 2017.

[4] W. Sun et al., "A simple and effective spectral–spatial method for mapping large-scale coastal wetlands using China ZY1–02D satellite hyperspectral images," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 104, Dec. 2021, Art. no. 102572.

[5] S. Jia, L. Shen, J. Zhu, and Q. Li, "A 3-D Gabor phase-based coding and matching framework for hyperspectral imagery classification," *IEEE Trans. Cybern.*, vol. 48, no. 4, pp. 1176–1188, Apr. 2018.

[6] G. Yang, K. Huang, W. Sun, X. Meng, D. Mao, and Y. Ge, "Enhanced mangrove vegetation index based on hyperspectral images for mapping mangrove," *ISPRS J. Photogramm. Remote Sens.*, vol. 189, pp. 236–254, Jul. 2022.

[7] J. Zhao, Y. Zhong, X. Hu, L. Wei, and L. Zhang, "A robust spectral–spatial approach to identifying heterogeneous crops using remote sensing imagery with high spectral and spatial resolutions," *Remote Sens. Environ.*, vol. 239, Mar. 2020, Art. no. 111605.

[8] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021.

[9] Y. Chen, X. Cao, Q. Zhao, D. Meng, and Z. Xu, "Denoising hyperspectral image with non-i.I.d. noise structure," *IEEE Trans. Cybern.*, vol. 48, no. 3, pp. 1054–1066, Mar. 2018.

[10] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[11] M. Wettle, V. E. Brando, and A. G. Dekker, "A methodology for retrieval of environmental noise equivalent spectra applied to four hyperion scenes of the same tropical coral reef," *Remote Sens. Environ.*, vol. 93, nos. 1–2, pp. 188–197, Oct. 2004.

[12] G. S. Okin and J. Gu, "The impact of atmospheric conditions and instrument noise on atmospheric correction and spectral mixture analysis of multispectral imagery," *Remote Sens. Environ.*, vol. 164, pp. 130–141, Jul. 2015.

[13] L. Zhuang, L. Gao, B. Zhang, X. Fu, and J. M. Bioucas-Dias, "Hyperspectral image denoising and anomaly detection based on low-rank and sparse representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2020.

[14] M. Wang, D. Hong, B. Zhang, L. Ren, J. Yao, and J. Chanussot, "Learning double subspace representation for joint hyperspectral anomaly detection and noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507517.

[15] J. Li, Q. Yuan, H. Shen, and L. Zhang, "Hyperspectral image recovery employing a multidimensional nonlocal total variation model," *Signal Process.*, vol. 111, pp. 230–248, Jun. 2015.

[16] W. Sun et al., "MLR-DBPFN: A multi-scale low rank deep back projection fusion network for anti-noise hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522914.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.

[17] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.

[18] J. Peng, Q. Xie, Q. Zhao, Y. Wang, L. Yee, and D. Meng, "Enhanced 3DTV regularization and its applications on HSI denoising and compressed sensing," *IEEE Trans. Image Process.*, vol. 29, pp. 7889–7903, 2020.

[19] L. Zhuang, M. K. Ng, and Y. Liu, "Cross-track illumination correction for hyperspectral pushbroom sensor images using low-rank and sparse representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502117.

[20] S. Jiang and X. Hao, "Hybrid Fourier-wavelet image denoising," *Electron. Lett.*, vol. 43, no. 20, pp. 1081–1082, Sep. 2007.

[21] B. Rasti, J. R. Sveinsson, M. O. Ulfarsson, and J. A. Benediktsson, "Hyperspectral image denoising using first order spectral roughness penalty in wavelet domain," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2458–2467, Jun. 2014.

[22] H. Fan, C. Li, Y. Guo, G. Kuang, and J. Ma, "Spatial–spectral total variation regularized low-rank tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6196–6213, Oct. 2018.

[23] Y. Chang, L. Yan, and S. Zhong, "Hyper-Laplacian regularized unidirectional low-rank tensor recovery for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4260–4268.

[24] T. Xie, S. Li, and B. Sun, "Hyperspectral images denoising via nonconvex regularized low-rank and sparse matrix decomposition," *IEEE Trans. Image Process.*, vol. 29, pp. 44–56, 2020.

[25] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.

[26] M. Wang, Q. Wang, J. Chanussot, and D. Li, "Hyperspectral image mixed noise removal based on multidirectional low-rank modeling and spatial–spectral total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 488–507, Jan. 2020.

[27] L. Zhuang, X. Fu, M. K. Ng, and J. M. Bioucas-Dias, "Hyperspectral image denoising based on global and nonlocal low-rank factorizations," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10438–10454, Dec. 2021.

[28] L. Zhuang and J. M. Bioucas-Dias, "Fast hyperspectral image denoising and inpainting based on low-rank and sparse representations," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 730–742, Mar. 2018.

[29] W. He, Q. Yao, C. Li, N. Yokoya, and Q. Zhao, "Non-local meets global: An integrated paradigm for hyperspectral denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6868–6877.

[30] M. Wang, Q. Wang, and J. Chanussot, "Tensor low-rank constraint and l_0 total variation for hyperspectral image mixed noise removal," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 718–733, Apr. 2021.

[31] Q. Xie et al., "Multispectral images denoising by intrinsic tensor sparsity regularization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1692–1700.

[32] Y. Chang, L. Yan, X. Zhao, H. Fang, Z. Zhang, and S. Zhong, "Weighted low-rank tensor recovery for hyperspectral image restoration," *IEEE Trans. Cybern.*, vol. 50, no. 11, pp. 4558–4572, Nov. 2020.

[33] J. Xue, Y. Zhao, W. Liao, and J. C. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.

[34] Y. Chen, W. He, N. Yokoya, T. Huang, and X. Zhao, "Nonlocal tensor-ring decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1348–1362, Feb. 2020.

[35] Q. Zhang, Q. Yuan, J. Li, X. Liu, H. Shen, and L. Zhang, "Hybrid noise removal in hyperspectral imagery with a spatial–spectral gradient network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7317–7329, Oct. 2019.

[36] Y. Chang, L. Yan, H. Fang, S. Zhong, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 667–682, Feb. 2019.

[37] Q. Zhang, Q. Yuan, J. Li, F. Sun, and L. Zhang, "Deep spatio-spectral Bayesian posterior for hyperspectral image non-i.I.d. noise removal," *ISPRS J. Photogramm. Remote Sens.*, vol. 164, pp. 125–137, Jun. 2020.

[38] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[39] K. Zhang, W. Zuo, and L. Zhang, "FFDNet: Toward a fast and flexible solution for CNN-based image denoising," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4608–4622, Sep. 2018.

[40] S. V. Venkatakrishnan, C. A. Bouman, and B. Wohlberg, "Plug-and-play priors for model based reconstruction," in *Proc. IEEE Global Conf. Signal Inf. Process.*, Dec. 2013, pp. 945–948.

[41] S. H. Chan, X. Wang, and O. A. Elgendy, "Plug-and-play ADMM for image restoration: Fixed-point convergence and applications," *IEEE Trans. Comput. Imag.*, vol. 3, no. 1, pp. 84–98, Mar. 2017.

[42] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[43] X. Fu, S. Jia, M. Xu, J. Zhou, and Q. Li, "Fusion of hyperspectral and multispectral images accounting for localized inter-image changes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517218.

[44] J. Zhou, W. Sun, X. Meng, G. Yang, K. Ren, and J. Peng, "Generalized linear spectral mixing model for spatial–temporal–spectral fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533216.

[45] L. Zhuang, C. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[46] L. Zhuang and M. K. Ng, "FastHyMix: Fast and parameter-free hyperspectral image mixed noise removal," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–15, 2021.

[47] L. Zhuang, M. K. Ng, and X. Fu, "Hyperspectral image mixed noise removal using subspace representation and deep CNN image prior," *Remote Sens.*, vol. 13, no. 20, p. 4098, Oct. 2021.

[48] Y. Zheng, T. Huang, X. Zhao, Y. Chen, and W. He, "Double-factor-regularized low-rank tensor factorization for mixed noise removal in hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8450–8464, Dec. 2020.

[49] Y. Chen, T. Huang, W. He, X. Zhao, H. Zhang, and J. Zeng, "Hyperspectral image denoising using factor group sparsity-regularized nonconvex low-rank approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515916.

[50] C. Cao, J. Yu, C. Zhou, K. Hu, F. Xiao, and X. Gao, "Hyperspectral image denoising via subspace-based nonlocal low-rank and sparse factorization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 3, pp. 973–988, Mar. 2019.

[51] Y. Romano, M. Elad, and P. Milanfar, "The little engine that could: Regularization by denoising (RED)," *SIAM J. Imag. Sci.*, vol. 10, no. 4, pp. 1804–1844, Jan. 2017.

[52] L. Zhuang and M. K. Ng, "Hyperspectral mixed noise removal by ℓ_1-norm-based subspace representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1143–1157, 2020.

[53] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.

[54] M. Simões, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[55] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Spectral–spatial hyperspectral image segmentation using subspace multinomial logistic regression and Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 809–823, Mar. 2012.

[56] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[57] J. Huang, T. Huang, L. Deng, and X. Zhao, "Joint-sparse-blocks and low-rank representation for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2419–2438, Apr. 2019.

[58] X. Fu, S. Jia, M. Xu, J. Zhou, and Q. Li, "Sparsity constrained fusion of hyperspectral and multispectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[59] Y. Chen, W. He, N. Yokoya, and T. Huang, "Hyperspectral image restoration using weighted group sparsity-regularized low-rank tensor decomposition," *IEEE Trans. Cybern.*, vol. 50, no. 8, pp. 3556–3570, Aug. 2020.

[60] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.



Xiyou Fu (Member, IEEE) received the bachelor's degree from Wuhan University, Wuhan, China, in 2012, and the M.S. and Ph.D. degrees from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2015 and 2019, respectively.

He is currently a Post-Doctoral Researcher with Shenzhen University, Shenzhen, China. His research interests include hyperspectral image restoration, anomaly detection, and super-resolution.



Yujuan Guo received the M.S. degree in geographic information systems from the Xi'an University of Science and Technology, Xi'an, China, in 2015, and the Ph.D. degree in geographic information systems from the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2021.

She is a Post-Doctoral Researcher with Shenzhen University, Shenzhen, China. Her research interests include hyperspectral image fusion, image segmentation, and deep learning.



Meng Xu (Member, IEEE) received the B.S. and M.E. degrees in electrical engineering from the Ocean University of China, Qingdao, China, in 2011 and 2013, respectively, and the Ph.D. degree from the University of New South Wales, Canberra, ACT, Australia, in 2017.

She is an Assistant Professor with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. Her research interests include cloud removal and remote sensing image processing.



Sen Jia (Senior Member, IEEE) received the B.S. and Ph.D. degrees from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2002 and 2007, respectively.

Since 2008, he has been with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China, where he is currently a Full Professor. His research interests include hyperspectral image processing, signal and image processing, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:45:52 UTC from IEEE Xplore. Restrictions apply.